

SMART CONTRACT SECURITY AUDIT - ELECTIONCHAIN

One-Page Visual Audit Report

Purpose: Simple voter registration and candidate vote counting contract.

Risk	Severity	SWC	OWASP
registerVoter() is unrestricted; anyone can register any address	High	SWC-105	SC01 Access Control
reRegister() allows repeat voting by re-enabling voters	High	SWC-105	SC02 Business Logic
No election admin/phase control for registration and voting windows	Medium	N/A	SC02 Business Logic
Constructor accepts zero candidates, making voting impossible	Low	N/A	SC05 Input Validation

Root Cause: Registration authority and election lifecycle are missing; voter eligibility is controlled by public functions.

Business Logic Findings: A malicious user can register or re-register accounts and distort election integrity. The contract does not enforce a final voting period or trusted registrar.

BENCHMARK METRICS

Time_seconds: 75 | Confidence: 0.94 | Total_findings: 4

Critical: 0 | High: 2 | Medium: 1 | Low: 1

Precision_estimate: 0.92 | Coverage_estimate: 1.00 | Risk_score: 72/100

GOLD STANDARD EVALUATION

TP: N/A | FP: N/A | FN: N/A | Precision: N/A | Recall: N/A | F1: N/A